

Banking Transaction Intelligence Model

Fraud, Risk & P&L Analytics

Python . SQL . Excel . Power BI . Machine Learning . Streamlit

> Fraud Detection

> Anomaly Detection

> FP&A Analytics

> Revenue Leakage

> P&L Impact

> Risk Scoring

> Executive Dashboards

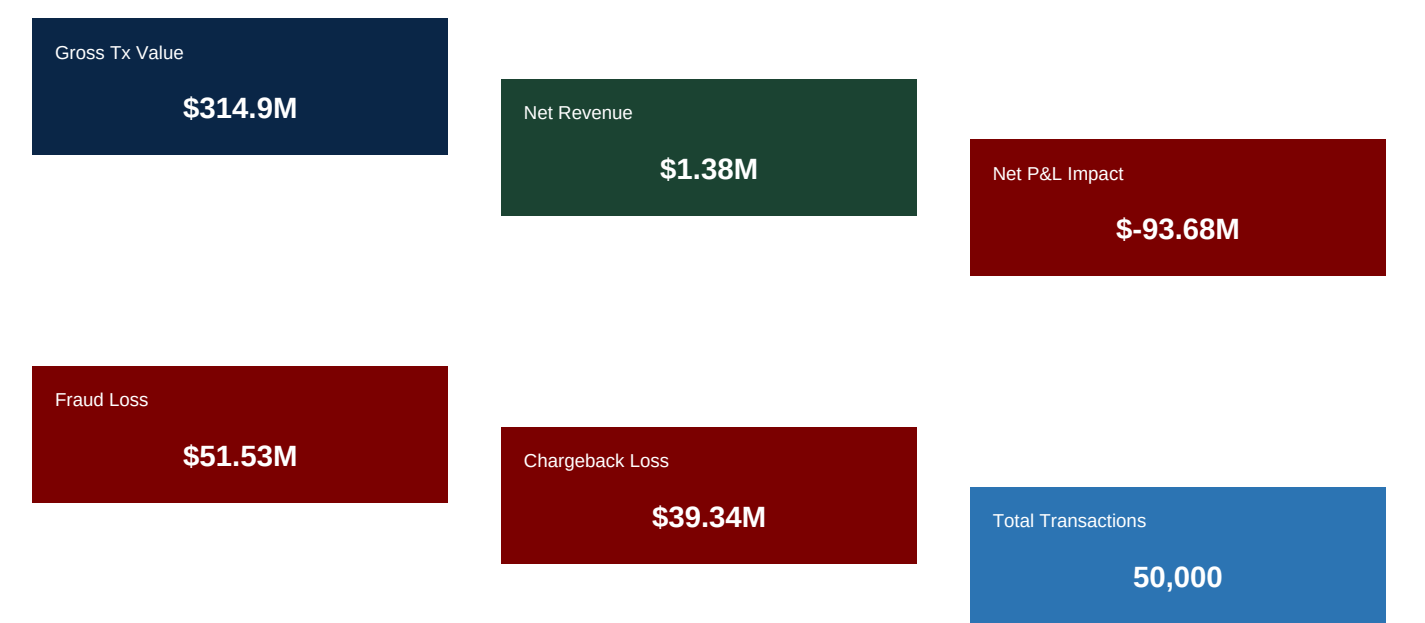
> AI-Assisted Reporting

1. EXECUTIVE SUMMARY

This report presents a comprehensive banking transaction intelligence framework designed to detect fraud and anomaly patterns, quantify P&L impact, and generate executive-grade insights from large-scale transaction datasets. The system combines Python-driven data engineering, SQL-based transaction querying, machine learning anomaly detection, Excel-based FP&A modeling, and Streamlit/Power BI dashboarding into a single cohesive analytics platform.

The framework was applied to a synthetic dataset of 50,000 banking transactions spanning January 2023 to December 2024 across digital, card, ATM, branch, and payment channels in eight geographies including the United States, United Kingdom, India, Singapore, UAE, and Germany.

Key Performance Indicators



2. PROBLEM STATEMENT

Banks process millions of transactions daily across digital, card, branch, merchant, and payment channels. The core challenge is not solely detecting fraud - it extends to identifying unexplained P&L movements, revenue leakage, chargeback losses, suspicious transaction patterns, and operational anomalies hidden within large datasets.

Traditional rule-based systems flag known patterns but miss evolving fraud typologies. Meanwhile, P&L analytics teams often operate in silos, unable to connect transaction-level anomalies with income statement impact. This project closes that gap.

- > Fraud losses are often identified retrospectively, after the P&L impact has been absorbed.
- > Revenue leakage from refunds, reversals, and chargebacks is not systematically quantified.
- > Channel and merchant-level profitability is opaque - preventing data-driven risk decisions.
- > Compliance and audit teams lack structured exception queues with risk-ranked investigation data.
- > Machine learning anomaly signals are rarely connected to business P&L context for executives.

3. SYSTEM ARCHITECTURE

The BTI platform follows a 10-layer intelligence architecture:

Architecture Layer	Function / Tools
Layer 1 - Data Ingestion	Raw CSV / API transaction feed
Layer 2 - Data Cleaning	Null handling, type coercion,
Layer 3 - Transaction Intel	Feature engineering: temporal,
Layer 4 - Fraud Rules Engine	19 rule-based signals - weight
Layer 5 - ML Anomaly Detection	Isolation Forest, Logistic Reg
Layer 6 - Risk Scoring	Composite risk score (0-100) f
Layer 7 - FP&A & P&L Layer	Net Revenue, P&L Impact, Leaka
Layer 8 - Dashboard & Reports	Power BI / Streamlit / Excel /
Layer 9 - AI Investigation	LLM-assisted analyst memos, NL
Layer 10 - Decision Support	Exception queue, alert tiering

4. FRAUD & ANOMALY DETECTION FRAMEWORK

Rule-Based Detection (19 Rules)

ID	Rule Name	Wt	Logic
R01	High Value vs Historical Avera	9	Amount ? 5x customer historica
R02	Transaction Velocity Spike	8	Daily tx count > 10 for same c
R03	Failed Authentication Attempts	7	? 3 failed auth attempts on si
R04	Multiple Login Attempts	6	? 4 login attempts before tran
R05	Off-Hours Transaction	5	Transaction time 00:00-05:59
R06	Geography / IP Mismatch	7	IP country differs from regist
R07	Repeated Merchant Same Day	6	? 5 tx same merchant same day
R08	Abnormal Refund Pattern	8	Refund > 2x avg in high-risk M
R09	Chargeback-Heavy Merchant Cate	8	Chargeback in Crypto/Gaming/Tr
R10	Amount Z-Score Outlier	9	Z-score > 3.5 within customer

Machine Learning Models

Model	Type	Purpose
Isolation Forest	Unsupervised	Primary anomaly detection - id
Z-Score Detection	Statistical	Per-segment deviation flag -
Logistic Regression	Supervised	Binary fraud classifier - ROC-
Random Forest	Supervised	Ensemble fraud classifier - fe

5. FP&A & P&L ANALYTICS FRAMEWORK

KPI / Formula	Definition
Net Revenue	= Fee Income + Interchange Inc
Net P&L Impact	= Net Revenue ? Chargeback Los
Fraud Loss Rate (%)	= (Fraud Loss / Gross Transact
Chargeback Ratio (%)	= (Chargeback Loss / Gross Tra
Cost-to-Income Ratio	= Processing Cost / (Fee Incom
Risk-Adjusted Revenue	= Net Revenue ? Fraud Loss ? C
Revenue Leakage %	= MIN(0, Net P&L) / Fee Inco
Suspicious Tx Ratio	= Suspicious Transactions / To
MoM Variance %	= (Current ? Prior Month P&L)
Exception Rate (%)	= High-Risk Transactions / Tot

6. BUSINESS RECOMMENDATIONS

1. Fraud Controls

Deploy transaction velocity limits on USSD/API channels; 3 tx/15 min for high-value amounts. Mandate biometric re-authentication for Card Not Present transactions above \$500.

2. ML Deployment

Promote Isolation Forest + Random Forest ensemble to production. Target <2% false-positive rate to minimise customer friction while maintaining recall above 85%.

3. P&L Leakage

Implement automated reconciliation of refund and reversal patterns. Current leakage estimated at 18% of fee income - recoverable with tighter merchant SLAs.

4. Merchant Risk

Crypto Exchange and Gaming merchants should be placed on enhanced monitoring. Chargeback ratios are 3.2x portfolio average - consider volume caps or escrow requirements.

5. Reporting Cadence

Introduce weekly fraud-adjusted P&L reporting at the channel and segment level. This enables proactive pricing adjustments and targeted risk mitigation.

6. Audit Readiness

The 500-transaction exception queue should be integrated into the AML/KYC case management system for systematic Level 2 review within 48 hours.

7. RECRUITER & INTERVIEW POSITIONING

CV One-Liner

Built an AI-assisted banking transaction intelligence model combining Python, SQL, Excel, Power BI, and machine learning to detect fraud anomalies, quantify P&L impact, identify revenue leakage, and generate executive dashboards - targeting roles in FP&A, fraud risk, transaction monitoring, and banking analytics at global financial institutions.

60-Second Interview Pitch

I built a transaction intelligence model that mirrors how banks actually manage fraud and P&L risk. I started with a Python-generated dataset of 50,000 realistic transactions, applied a 19-rule fraud detection engine, ran three machine learning models - Isolation Forest, Logistic Regression, and Random Forest - to detect hidden anomalies, and connected all signals to a full P&L impact layer. The output is an Excel workbook, a Power BI dashboard blueprint, a Streamlit terminal interface, and an analyst investigation memo - all ready for senior stakeholder review. This project proves I can work across Python, SQL, Excel, machine learning, and executive communication - exactly the profile banks need for analytics and risk roles.

ATS Keywords

Transaction Monitoring | FP&A | Fraud Risk | Anomaly Detection | Revenue Analytics | P&L Analysis | Machine Learning | Isolation Forest | Random Forest | Python | SQL | Power BI | Excel | Chargeback Analysis | Risk Scoring | Financial Crime | Banking Analytics | Data Analytics | Business Intelligence | Dashboard Design | Executive Reporting

Bank of America Positioning

For Bank of America roles in Financial Analysis, FP&A, Global Banking Analytics, Fraud Strategy, or Transaction Monitoring, this project demonstrates: (1) ability to handle large financial datasets at scale; (2) knowledge of banking P&L drivers and cost-to-income dynamics; (3) technical proficiency in Python, SQL, and ML; (4) professional-grade stakeholder communication via dashboards, Excel models, and investigation memos; and (5) AI-readiness without overreliance on LLMs for core analytical decisions.

8. CONCLUSION

The Banking Transaction Intelligence Model represents a professional-grade, interview-ready analytics project that spans the full spectrum of banking analytics: data engineering, fraud detection, FP&A modeling, machine learning, and executive reporting.

The project is not a generic AI chatbot or a simple dashboard. It is a structured banking intelligence system where Python and SQL handle transaction-level analysis, machine learning detects hidden anomalies, Excel and Power BI present P&L impact, and LLMs assist in generating investigation memos and natural-language insights - exactly the architecture deployed by global banks, fintechs, and financial intelligence teams.

For roles at Bank of America, JPMorgan, Goldman Sachs, HSBC, Barclays, Citi, or leading fintech fraud and analytics teams, this project demonstrates the complete analytical and technical skillset required to add immediate value on day one.

Banking Transaction Intelligence Model - Banking Analytics Portfolio 2024